

P15 Daily, money and GST reports

Model: Opus, high effort.

Branch: none. Commit directly to `main`.

Depends on: P14.

1. What to build, in one sentence

Add nine report definitions to the engine from P14: R2 Day Close, R3 Sales by Day, R4 Hours and Weekdays, R5 Payments, R6 Platform Money, R7 Cash Till, R8 GST, R9 Tally Export and R10 Invoice Register, each proven against the golden day.

2. Module

M19 Reports v2. Server only. The screens come in P18.

3. The spec is already written

`docs/API-CONTRACT.md` section "M19 Reports v2", from P13, gives each report's endpoint, roles, filters, columns, stored fields, totals, checks and drill downs.

`docs/REPORT-SPEC.md` says what each report is for.

Build exactly what the contract says. If it and this prompt disagree, the contract wins, and you tell me.

If the contract is silent on something you need, stop and ask.

4. Step 0. Save this prompt and check the repo

1. Save this entire prompt, exactly as given, to `docs/prompts/P15-daily-money-gst-reports.md`.
2. `git pull` . `git status` should show only that file.
3. `docs/prompts/README.md` must show P14 as Done. If not, stop.
4. Run `npm test` and record the count. It should match P14's "after" count. If anything fails, stop and tell me.

5. Files to read first

1. The M19 contract entries for R2 to R10.
2. `docs/REPORT-SPEC.md` R2 to R10, and `docs/GLOSSARY.md`.
3. `docs/TEST-DATA.md` sections 4, 5 and 6.
4. `server/services/reports/engine.js` , `registry.js` , `labels.js` , and `definitions/bills.js` from P14, as the pattern.
5. `server/services/dayFiguresService.js` and `server/services/reconciliationService.js`.
6. `server/services/payoutService.js` from P09.
7. `server/utills/tax.js` , especially `largestRemainderSplit`.
8. The existing M6 services and tests for sales by day and hourly, since R3 and R4 may extend their paths.

6. The nine definitions

One file each in `server/services/reports/definitions/` , registered in `registry.js` .

Every definition follows the rules at the top of `engine.js` : frozen fields only, paise only, averages after totalling, no reads of menu, categories, users or payment methods.

Report	Main source	Points to get right
R2 Day Close	<code>computeDayFigures</code> for an open day. The stored	Never a second calculation. For a closed day, C12 compares the

Report	Main source	Points to get right
	<code>dayclosures.snapshot</code> for a closed day.	snapshot with a fresh computation.
R3 Sales by Day	Bills by <code>businessDate</code>	Every date in the range gets a row, zeros where there were no bills. The "previous range" comparison, if the contract includes it, uses a range of the same length ending the day before <code>from</code> .
R4 Hours and Weekdays	Bills by the hour of <code>billedAt</code> in <code>DISPLAY_TIMEZONE</code>	All 24 hours present. The weekday grid has 7 rows, Monday first, each with 24 cells.
R5 Payments	Payments by their own <code>businessDate</code>	Every active method, and every method used in the range even if now inactive, gets a column, zeros where unused. On Hold from <code>chargedToAccountInPaise</code> . Collections in their own section. OWNER only.
R6 Platform Money	<code>platformpayouts</code> and platform payments	One row per live payout batch with expected through <code>payoutService.expectedFor</code> , plus a section of platform payments no live payout covers. Rate-not-set payments listed, never estimated.
R7 Cash Till	<code>dayclosures</code> and <code>computeDayFigures</code>	One row per business date. Closed days from the stored close. Open days show expected cash so far and no count. OWNER only.

Report	Main source	Points to get right
R8 GST	Bills' <code>taxBreakdown</code> and <code>taxTreatment</code>	Section B, platform-collected, by frozen <code>platform.code</code> . Section C, No Charge value and voided totals, never added into section A. Section D, documents per <code>invoiceSeries</code> . Section E, round-off.
R9 Tally Export	As R8, plus payments	A file only. The exact sheet layout from the contract. See section 7 below. Refuses to build if any ERROR check fails.
R10 Invoice Register	Every bill, voided included, ordered by series then sequence	A gap shows as its own "Missing number" row. A null <code>invoiceSeries</code> belongs to its <code>financialYear</code> series.

Where the contract keeps an existing M6 path, as `docs/REPORT-SPEC.md` section 4 says for R3 and R4, the existing M6 tests on that path must still pass unchanged, unless the contract deliberately changes a field. If it does, change those tests on purpose and list each one in your summary.

7. R9: splitting bills across payments

The Tally file has two blocks: sales by tax rate, and sales by payment method, with each method's Tally code.

The by-rate block is simple: each rate's net sales, CGST and SGST, straight from the bills' `taxBreakdown` , and round-off on its own row.

The by-method block needs each bill's net sales and GST divided between the ways it was paid.

Caffeza's old export worked the taxable value backwards from the rounded amount, which is why its two blocks show different taxable

totals for the same day.

Ours must not.

1. Add `splitBillAcrossPayments(bill)` to `server/utils/tax.js`, the only place tax arithmetic lives.
2. For one bill, the parts are its payments, plus its `chargedToAccountInPaise` as an On Hold part, plus any unpaid remainder as an Unpaid part.
3. Split the bill's net sales across the parts with `largestRemainderSplit`, weighted by each part's amount. Split its CGST and SGST the same way, separately. The round-off stays on its own row and is not split.
4. Before returning, check that each split adds back exactly to the bill's own figure, and throw if not, the same discipline as `allocateLineShares`.
5. The by-method block then adds these parts up by frozen method code, with the On Hold part on its own row using Caffeza's `P03` code from the On Hold setting or the contract.

The two blocks must have exactly the same total net sales, CGST and SGST. Test it.

8. Checks

Each report runs the checks the contract lists for it, through the engine. R9 refuses to build when any ERROR check fails, and says which.

9. Tests

Build the golden day once per test file with `tests/helpers/goldenDay.js`, close 26 September as in P10, and check each report against `docs/TEST-DATA.md`.

R2

1. For 26 September, every section matches docs/TEST-DATA.md section 4, to the paisa.
2. For the closed day, the response is the stored snapshot, and C12 passes. After changing B06's stored total directly, C12 fails.

R3

3. One row for 26 September: 15 bills, 27 covers, item total ₹9,310.22, discount ₹423.90, net sales ₹8,886.32, GST ₹382.57, round-off ₹0.11, bill total ₹9,269.00, average bill ₹592.42, average per cover ₹267.09.
4. With the section 5 scenario added for 27 September, a range of 26 to 28 September has three rows, 28 September all zeros, and openDays lists 27 and 28 September.

R4

5. Net sales by hour of billing, matching:

Hour	Bills	Net sales
12	1	₹477.00
13	1	₹1,376.93
14	2	₹1,360.00
15	1	₹757.61
16	1	₹717.61
17	1	₹930.00
18	2	₹350.00
19	1	₹480.00
20	2	₹1,071.00
21	2	₹840.00
23	1	₹526.17

Every other hour is zero, and the hours add up to ₹8,886.32.

R5

6. 26 September: Cash ₹1,754.00, Card ₹1,481.00, UPI ₹1,472.00, money in hand ₹4,707.00. Zomato Gold ₹1,998.00, Dineout ₹778.00, EazyDiner ₹0.00, Zomato ₹305.00, Swiggy ₹930.00, platform money ₹4,011.00. On Hold ₹551.00. Unpaid ₹0.00. Total ₹9,269.00.

7. B14's payment at 12:02 AM on 27 September is counted on 26 September.

8. With the section 5 collection added, 27 September's collections section shows W-330 Office, cash, ₹504.00, and 27 September's sales figures do not include it.

9. A MANAGER is refused.

R6

10. With Swiggy at 2000 basis points and a payout for 26 September of ₹744.00: expected ₹744.00, difference ₹0.00. Zomato Gold, with no rate set, appears under "rate not set".

R7

11. 26 September: opening float ₹2,000.00, cash from bills ₹1,754.00, collections ₹0.00, paid in ₹0.00, paid out ₹350.00, expected ₹3,404.00, counted ₹3,400.00, difference -₹4.00, closed by Manager.

R8

12. Section A: 5%, net sales ₹7,651.32, CGST ₹191.31, SGST ₹191.26. Section B: Swiggy ₹930.00 and Zomato ₹305.00, totalling ₹1,235.00. Section C: No Charge ₹230.00 and voided ₹347.00. Section D: series CFA/C/ , first 22442, last 22457, 16 issued, 1 voided. Section E: round-off ₹0.11.

R9

13. The by-rate block: 0% ₹1,235.00. 5% taxable ₹7,651.32, CGST ₹191.31, SGST ₹191.26, final ₹8,033.89. Round-off ₹0.11. Total ₹9,269.00.

14. The by-method block totals exactly the same net sales, CGST and SGST as the by-rate block.

15. B05, paid ₹500.00 cash and ₹295.00 UPI on ₹757.61 net and ₹37.88 GST, splits so that the two parts add back exactly to ₹757.61 and

₹37.88.

16. Unit tests for `splitBillAcrossPayments` : a bill with one payment, a split payment, a partly paid bill charged to an account, a 0% bill, and a property test over 1,000 seeded random bills where every split adds back exactly.

17. With one ERROR check broken, the export refuses and names the check.

18. Open the file with `exceljs` and check every sheet's layout and Tally codes.

R10

19. 16 rows, CFA/C/22442 to CFA/C/22457, with B11's number marked voided, "Billed to the wrong table", B09 and B10 marked On Hold, the rest Paid, and no missing numbers.

20. Deleting B11's stored document directly makes a "Missing number" row appear for CFA/C/22452, and C6 fails.

Every report

21. Each one's `xlsx` export opens and its totals equal the JSON response.

22. Each one's roles, from the contract's permission table.

Run the full suite at the end. Every test that passed before must still pass.

10. Non-negotiable rules that apply

"Reports only add up values frozen onto records when the event happened. A report never recomputes tax."

"Tax arithmetic lives in `server/utils/tax.js` . Nowhere else." The payment split lives there.

"Every report goes through the one shared report engine."

"Every label on a screen or an export comes from `docs/GLOSSARY.md` ."

"A totals row is the exact sum of its rows. An average is a sum divided by a sum."

"Store timestamps in UTC. Convert to India time only for display." R4 converts hours in the aggregation, using `DISPLAY_TIMEZONE` .

11. Checks and golden day

Every report here is proven against the golden day, to the paisa.

If a number differs from `docs/TEST-DATA.md`, do not change the expected number. Find which side is wrong and tell me.

12. Docs to update

1. `docs/API-CONTRACT.md` M19: anything the build had to settle that the contract left open, especially R9's on-hold Tally code and the payment split rule.
 2. `docs/PROJECT-STATE.md` :
 1. The date line.
 2. "Current stage": "Next: P16, menu, captain and table reports."
 3. Decision log, dated today: "The Tally by-method block splits each bill's net sales, CGST and SGST across its payments and account charge with the largest remainder method, in `splitBillAcrossPayments` in `tax.js`. Both Tally blocks always total the same. | Caffeza's old export worked figures backwards from rounded amounts and its two blocks disagreed."
 4. Every place the build differed from the contract, as a decision row, and the contract updated to match.
 5. "What changed recently": a P15 entry at the top, and the oldest moved to the archive.
 3. `docs/prompts/README.md` : mark P15 as Done.
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13. Out of scope

Screens and charts. That is P18.

R1 Today. That is P18.

R11 to R18. That is P16 and P17.

Tally XML vouchers. Excel only for now.

Changing or removing any M6 endpoint beyond what the contract says.

14. Done when

1. `npm test` passes, with before and after counts recorded.
2. `npm run lint` and `npm run build` pass.
3. By hand, with `seed:demo` data or a staging-like setup: call each of the nine endpoints for a recent date and download R9 and R10 as Excel.
4. Every doc in section 12 is updated.
5. Commits on `main`, one line each, for example:
 - `add day close sales by day and hours reports`
 - `add payments platform money and cash till reports`
 - `split bills across payments in tax`
 - `add gst tally and invoice register reports`
 - `update docs for p15`
6. Push `main`.
7. Print a short summary: commits, files changed, test counts before and after, and every place the build differed from the contract.